

How to describe the spatial near-far relations among concepts?



Keyin Zheng^{a,c}, Yuhua Qian^{a,b,c,*}, Honghong Cheng^{d,a}

^a Institute of Big Data Science and Industry, Shanxi University, Taiyuan, 030006 Shanxi, China

^b Key Laboratory of Computational Intelligence and Chinese Information Processing of Ministry of Education, Shanxi University, Taiyuan, 030006 Shanxi, China

^c Engineering Research Center for Machine Vision and Data Mining of Shanxi Province, Shanxi University, Taiyuan 030006, Shanxi, China

^d School of Information, Shanxi University of Finance and Economics, Taiyuan, 030012 Shanxi, China

ARTICLE INFO

Article history:

Received 8 March 2022

Received in revised form 8 February 2023

Accepted 9 February 2023

Available online 28 February 2023

Keywords:

Granular computing

Concept

Relation

Attribute-based

Object-based

Salient representation

ABSTRACT

Research in cognitive science shows that when humans learn concepts, relation-based categories are easier to learn than feature-based categories, and humans can grasp the essential difference of things in different relations and distinguish a certain degree of diversity. For example, in terms of color, blue sea is closer to blue sky than red coral. How can the machine understand the near-far relations among different concepts? Drawing on the idea of granular computing and starting from the perspective of relation, this study proposes a method to describe the spatial near-far relations among different concepts. First, the eigenvalue representation methods of the matrix are given, and on this basis, the salient feature space of the symmetric matrix is proposed. Then, we propose a salient representation method for category, and through qualitative analysis from the perspective of consistency and near-far relations, we find that the salient representation can represent categories from the perspective of attribute and object respectively. Finally, we verify the validity of salient relation representation through second-order isomorphism between intent and extent of concept quantitatively.

© 2023 Elsevier Inc. All rights reserved.

1. Introduction

The relations of concepts play a fundamental role in people's reasoning, judgment, and understanding of the world around them, how to automatically discover, learn, and apply the relations of concepts to recognize objects, make connections, discover patterns, and exchange ideas like human beings, has been an important topic in cognitive science [17,25,24,30,29,56,57], machine learning [30,38,62,78], artificial intelligence, and other fields. Different categories of the same thing can be understood as concretization concepts that can reflect the existence of things under different conditions. Research in cognitive science shows that when humans learn concepts, relation-based categories are easier to learn than feature-based categories [59], and humans can grasp the essential difference of things in different relations and distinguish a certain degree of diversity. For example, in terms of color, blue sea is closer to blue sky than red coral. Many real-world concepts have a strong relational foundation [15], providing more information for machines to explore concepts and their

* Corresponding author at: Institute of Big Data Science and Industry, Shanxi University, Taiyuan, 030006 Shanxi, China.

E-mail addresses: zhengkeyin1221@163.com (K. Zheng), jinchengqyh@126.com (Y. Qian), chhsxdx@163.com (H. Cheng).

relations, and complete higher-order tasks. How the machine can understand the near-far relations among different concepts is crucial for concept learning.

The purpose of this study is to consider the near-far relations from the perspective of inter-category. From the perspective of methodology, we consider the structure of different intra-category relations and then obtain inter-category near-far relations. Therefore, we introduced them from these two perspectives.

For inter-category relations, different areas have produced different understandings and researches.

1) *Granular computing perspective*. Granular computing is a common paradigm for describing and dealing with problems from multiple perspectives and at multiple levels. It simulates human cognition of the world and reasoning to solve problems by simplifying the solution of complex problems through information granulation [3,42,44,63,65,72,74–76]. A granule is the most basic element of a granular computing model. It can be understood as a concept, and an element in a granule can be understood as an instance of a corresponding concept. As early as 1982, the German mathematician Wille formalized the concept of philosophy as a formal concept (concept lattice) [14,70]. The concept lattice structure model is used to describe the relationship between objects (samples) and attributes (features), where all concept structures are constructed according to certain partial-order relations and the definitions of the upper and lower definite bounds [69]. On this basis, the methods of concept and their relations emerge from the cognitive perspective—concept-cognitive learning model [21,32–34,39,41,49,50,64,73,77]. Inspired by philosophy and cognitive psychology, Li et al. [34] proposed object-based and attribute-based cognitive granular concepts respectively, and the concept learning of additional information is realized through dynamic incremental association. An axiomatic method of forming three-way cognitive concepts based on multi-granularity and three-way-decision-making principles was firstly proposed in [32]. In [21,41] the authors focused on the relationship between the global and local granular concepts. Three-way concept learning was reconsidered based on cognitive operators from the perspective of information fusion in [21]. Shrivhare et al. [51] established cognitive relation between the pair of objects and attributes using the relational attribute and object method. Researchers can learn specific rule knowledge through the relations between concepts [73]. In many cases, these relations cannot be subdivided. For example, objects with common attributes usually have several categories, and we cannot describe the near-far relations among these categories using the conceptual association method described above.

2) *Learning perspective*. Since the beginning of machine learning, methods for learning concepts with better generalization have been explored. Most tasks in the machine learning field involve making machines understand and learn concepts better. For example, regression and classification map the sample data to a pre-defined label space by learning a classifier [5,18]. A concept is a mapping from the sample space to the label space that determines the true label of the sample. Most studies use category symbols or labels to represent concepts. The set of all the target concepts that we want to learn is called concept classes [1,4,12,40,61]. Clustering divides a data set into different clusters according to a specific standard (such as the distance between samples), so as to make the similarity of data objects in the same cluster as much as possible, and the difference of data objects not in the same cluster as much as possible. A cluster can be understood as a concept, and a prototype or cluster center can be found to represent the concept. Generally speaking, concept is the abstraction of category, which can be understood as a group of entities of a concept and is a concrete representation of the concept, the essence of the methods mentioned above is to describe categories from samples to achieve machine learning tasks. However, the description and understanding of the relation among categories are not intuitive. The supervised learning paradigm in machine learning represents a category using category label and samples belonging to this label [6,7,10,13,28,30]; however, labels are represented with no distance distinction, whereas the unsupervised machine learning paradigm makes the distance within categories closer and the distance among categories further by constructing similarity metrics, regardless of what the category is. Although there is a concept of distance among categories, owing to the goal of longer distances among categories, it is better to separate the different categories without considering the actual relations among categories [2,9,20,35]. The basis of the human cognitive process is the similarity among things, similar things usually behave similarly [16,60,45], and we can use these similarities to build relations among different concepts.

For intra-category relations, there are emerging works attempting to study the relation structure among different variables to achieve higher-order tasks, such as classification [36], recognition [71,66], attribute prediction [22,71,66], retrieval [26,37], and reasoning [46,23]. There is a great deal of research on concepts that use information entropy to determine the relative importance of attributes or among them. As an important measure of knowledge uncertainty and knowledge reduction, the entropy theory developed by Shannon [47] was used to obtain the precedence order of attributes for concept compression [55], reduce the concept lattice and fuzzy concept lattice according to different requirements [31,54,53] in formal concept analysis, and measure the uncertainty of the classification results. To deal with the large number of formal concepts and storage complexity from ‘a large context’, Li et al. [31] proposed an entropy-based weighted method to effectively reduce the general concept lattice based on information entropy and deviance analysis. Singh [54] proposed a method based on Shannon entropy and Huffman coding, where near-far reduction in fuzzy concepts was defined. Wang et al. [67] used mutual information [58] to define the strength of mutual dependence among attributes, and then selected attribute pairs with strong dependence to build an attribute relation graph. In [27], mutual information was used to describe the relations among attributes and an entropy reduction criterion was defined to predict the influence of new labels. These methods either directly use constructed relations or perform certain reductions based on expert knowledge to achieve higher-order tasks, such as knowledge discovery and image classification. However, the research [19] showed that some useless variables are not really redundant taking Gaussian

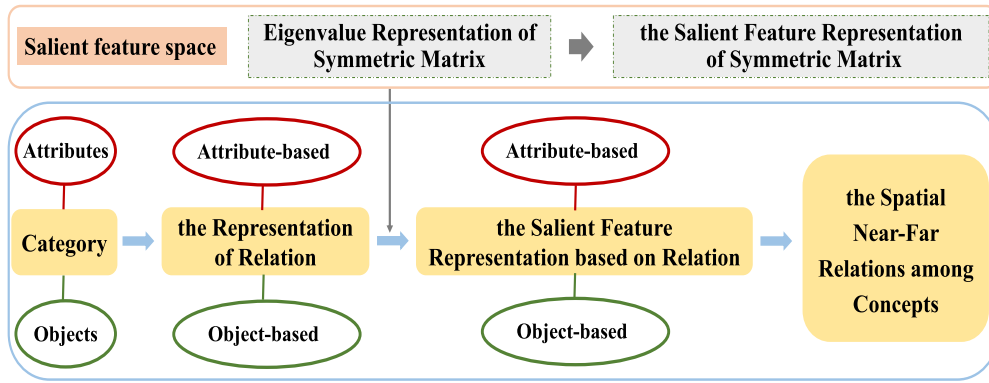


Fig. 1. The salient feature representation framework.

distribution as an example. A variable that is useless by itself can significantly improve the performance when used with other variables, and two variables that are useless still provide good class separability when used together. Instead of reducing, the method presented in this article relates all variables and uses all relation information for research on the structure itself represented by the relations and the effect of the structure on near-far relations among concepts.

Based on these premises, this study aims to address these problems. 1) How can we find a representation that describes the near-far relations among different concretization concepts? 2) How can we evaluate the near-far relations among the concepts we describe?

Symmetric matrices are often used to express specific relations between variables and are widely used. Research on symmetric matrices will help us better understand and extract useful relational structures. Based on the system theory, the whole is greater than the sum of its parts, the whole is composed of parts, and each part of the whole is combined in a certain structural way, has its own function, and produces the effect of $1 + 1 > 2$. Tao et al. [11] successively interpreted a method for solving the eigenvectors from eigenvalues from 2019 to 2021. Inspired by the idea of granular computing, we explore the near-far relations among concepts, mainly from the perspective of the structure of their internal relations. Starting with the eigenvalue representation of a symmetric matrix, we proposed a salient feature space. Then, based on attribute relation and object relation, the attribute-based salient category representation and object-based salient category representation are respectively given using the salient feature space. The consistency of relation representation and second-order isomorphism between internal and external representations show that this representation is effective for describing spatial near-far relations among concepts. Fig. 1 shows the primary framework of the proposed method.

The main contributions of the proposed method are summarized as follows.

- We propose a salient feature space based on the eigenvalue representation of a symmetric matrix. (Section 2)
- We provide a method for the salient relation representation of categories that can distinguish different concepts and describe the spatial near-far relations among concepts. (Section 3)
- We propose second-order isomorphism, consistency, and near-far relations to qualitatively and quantitatively demonstrate the effectiveness of the salient relation representation. (Subsection 3.3)

The remainder of this study is organized as follows: Section 2 introduces eigenvalue representation of matrix, including diagonalizable matrix, normal matrix and real symmetric matrix, and on this basis, the salient feature space is proposed. Section 3 details the relation-based category representation. In Section 4, we conduct a series of experiments on real datasets and present the performance of the proposed method. Finally, we draw conclusions in Section 5.

2. Salient feature space

Relations among variables can often be represented by symmetric matrices such as the adjacency matrices of undirected graphs. Real symmetric matrices are widely used in many scientific problems, and are of significant theoretical and practical value for studying the decomposition and properties of symmetric matrices. In this section, the salient feature space of a symmetric matrix is defined based on the eigenvalue representation of the symmetric matrix. Tao et al. [11] provided the eigenvalue representation of the matrix eigenvector, according to which, from the most general diagonalizable matrix, the eigenvalue representation methods of three kinds of matrices are given successively, including the diagonalizable matrix, normal matrix, and symmetric matrix. We define the salient feature space of the symmetric matrix on the basis of the eigenvalue representation of the matrix, and compare it with the original feature space.

2.1. Eigenvalue representation of matrix

We introduce several intermediate results in Tao's proof of eigenvector-eigenvalue-identity using adjugate matrices [11] in this subsection, and represent the diagonalizable matrix, normal matrix and real symmetric matrix with their eigenvalues respectively.

A diagonalizable matrix M_D is a $n \times n$ matrix with some eigenvalues $\lambda_1(M_D), \lambda_2(M_D), \dots, \lambda_n(M_D)$, which are not necessarily real or distinct, unless otherwise specified, the eigenvalues all satisfy the order relation from large to small, that is, $\lambda_1(M_D) \geq \lambda_2(M_D) \geq \dots \geq \lambda_n(M_D)$. Diagonalizable matrix M_D can be diagonalized, then M_D is similar to the diagonal matrix, and the elements on the main diagonal of the diagonal matrix that is similar to it are all the eigenvalues of M_D . M_D can be expressed as

$$M_D = \sum_{z=1}^n \lambda_z(M_D) v_z w_z^T, \quad (1)$$

where $\lambda_z(M_D)$ is the z -th eigenvalue of matrix M_D , $z = 1, 2, \dots, n$, v_z is the z -th right eigenvector of M_D , and w_z is the z -th left eigenvector of M_D , w_z^T is the transpose of w_z , that is $M_D v_z = \lambda_z(M_D) v_z$, and $w_z^T M_D = w_z^T \lambda_z(M_D)$. If v_z is an eigenvector of M_D with the eigenvalue $\lambda_z \neq 0$, then v_z is also an eigenvector of $\text{adj}(M_D)$ with the eigenvalue $\det(M_D)/\lambda_z$, where $\text{adj}(M_D)$ represents the adjoint matrix of M_D .

According to [11], $v_z w_z^T$ can be expressed as

$$\text{adj}(\lambda_z(M_D)I_n - M_D) = \left(\prod_{k=1; k \neq z}^n (\lambda_z(M_D) - \lambda_k(M_D)) \right) v_z w_z^T. \quad (2)$$

One easily knows that

$$M_D = \sum_{z=1}^n \lambda_z(M_D) \frac{\text{adj}(\lambda_z(M_D)I_n - M_D)}{\prod_{k=1; k \neq z}^n (\lambda_z(M_D) - \lambda_k(M_D))}. \quad (3)$$

Any normal matrix can be transformed into a diagonal matrix by a unitary transformation. In Eq. (1), in the case when matrices M_D is a normal matrix, and v_z are unit eigenvectors, then v_z and w_z^T are each other's conjugate transpose, that is $w_z^T = v_z^*$. It follows that a normal matrix M_N can be expressed as

$$M_N = \sum_{z=1}^n \lambda_z(M_N) v_z v_z^*, \quad (4)$$

$$\text{adj}(\lambda_z(M_N)I_n - M_N) = \left(\prod_{k=1; k \neq z}^n (\lambda_z(M_N) - \lambda_k(M_N)) \right) v_z v_z^*, \quad (5)$$

where $\lambda_z(M_N)$ is the z -th eigenvalue of matrix M_N , v_z is the z -th right eigenvector of M_N , v_z^* is the conjugate transpose of v_z . Then from Eq. (4), (5), one easily knows that

$$M_N = \sum_{z=1}^n \lambda_z(M_N) \frac{\text{adj}(\lambda_z(M_N)I_n - M_N)}{\prod_{k=1; k \neq z}^n (\lambda_z(M_N) - \lambda_k(M_N))}. \quad (6)$$

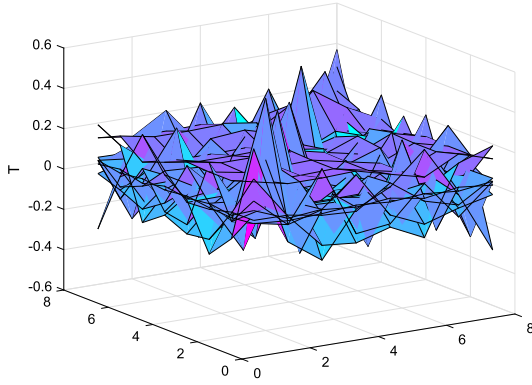
A real symmetric matrix must be a hermitian matrix, and a hermitian matrix must be a normal matrix. In particular, a real symmetric matrix is orthogonally similar to a real diagonal matrix; therefore, a real symmetric matrix must be diagonalized. The eigenvalues of a real symmetric matrix are real numbers, the eigenvectors of a real symmetric matrix are real vectors, and the eigenvectors corresponding to the different eigenvalues are orthogonal. The following discussion focuses on real symmetric matrices. For $n \times n$ real symmetric matrices M_S , we have

$$M_S = \sum_{z=1}^n \lambda_z(M_S) \frac{\text{adj}(\lambda_z(M_S)I_n - M_S)}{\prod_{k=1; k \neq z}^n (\lambda_z(M_S) - \lambda_k(M_S))}. \quad (7)$$

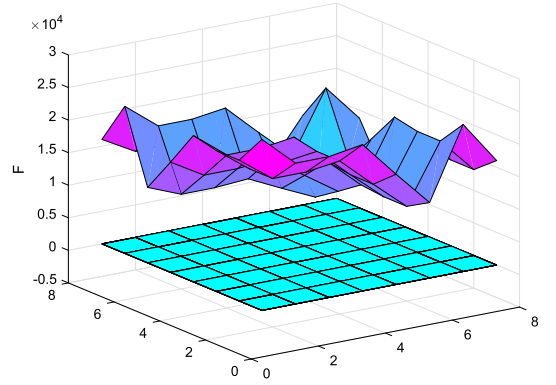
2.2. Salient feature representation of real symmetric matrix

In Eq. (7), symmetric matrix M_S can be represented as the sum of the product of the corresponding elements of an eigenvalue vector $\lambda_{n \times 1}$ and a tensor $T_{n \times n \times n}$, that is

$$M_S = \lambda_{n \times 1} \otimes T_{n \times n \times n}, \quad (8)$$



(a) Original feature space



(b) Salient feature space

Fig. 2. Difference between original feature space and salient feature space.

where the eigenvalue vector is expressed as $\lambda_{n \times 1} = [\lambda_1, \lambda_2, \dots, \lambda_n]^T$, and the z th component of tensor T is expressed as $T_z = T(:, :, z) = \frac{\text{adj}(\lambda_z(M_S)I_n - M_S)}{\prod_{k=1; k \neq z}^n (\lambda_z(M_S) - \lambda_k(M_S))}$. For convenience, we make $T_i = T(:, :, i)$ and $T_j = T(:, :, j)$, where $i, j = 1, 2, \dots, n$. Any columns in T_i are orthogonal to any columns in T_j , then we use T to represent the original feature space of the symmetric matrix M_S .

The symmetric matrix M_S can also be represented as the sum of the product of the corresponding elements of eigenvalues $\theta_{n \times 1}$ and a tensor $F_{n \times n \times n}$, that is

$$M_S = \theta_{n \times 1} \otimes F_{n \times n \times n}, \quad (9)$$

where $\theta_{n \times 1} = [(\prod_{k=1; k \neq 1}^n (\lambda_1(M_S) - \lambda_k(M_S)))^{-1}, (\prod_{k=1; k \neq 2}^n (\lambda_2(M_S) - \lambda_k(M_S)))^{-1}, \dots, (\prod_{k=1; k \neq n}^n (\lambda_n(M_S) - \lambda_k(M_S)))^{-1}]^T$, and the z th component of tensor F is expressed as $F_z = \lambda_z(M_S) \text{adj}(\lambda_z(M_S)I_n - M_S)$. Similar to above, for convenience, we make $F_i = F(:, :, i)$ and $F_j = F(:, :, j)$, where $i, j = 1, 2, \dots, n$. Any columns in F_i are orthogonal to any columns in F_j , then F is used to represent the salient feature space of the symmetric matrix M_S . It should be pointed out here that the different feature layers of the matrix are orthogonal to each other, so they can be added, the sum of all the components of the tensor F can be expressed as $\mathcal{F}(M_S)$, which is called salient representation.

$$\mathcal{F}(M_S) = \sum_{z=1}^n \lambda_z(M_S) \text{adj}(\lambda_z(M_S)I_n - M_S). \quad (10)$$

And we have that

$$\text{sum}(\theta_{n \times 1}) = \sum_{z=1}^n \frac{1}{\prod_{k=1; k \neq z}^n (\lambda_z(M_S) - \lambda_k(M_S))} = 0 (n \geq 2). \quad (11)$$

If M_S is a real $n \times n$ symmetric matrix, $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$, then $\lambda_1 I - M_S \geq 0$.

Generally, eigenvectors are used in Eq. (1) to describe the feature space of a matrix, or we can use the eigentensor T in Eq. (8) to represent the feature space of the matrix. In this section, we propose the concept of salient feature space F , and we visualize and compare the feature spaces represented by tensors T and F obtained by the same real symmetric matrix. The tensor represented by T is called the original feature space, and the tensor represented by F is called the salient feature space.

Fig. 2 shows the difference between the original feature space and salient feature space with the same symmetric matrix. In Fig. 2(a), we visualized the tensor T decomposed in Eq. (8), as shown in Fig. 2(a), each layer feature in the feature space described by the tensor T cannot be distinguished. But in Fig. 2(b), we visualized the tensor F decomposed in Eq. (9), in the feature space represented by the tensor F , the feature layer corresponding to the maximum eigenvalue can be separated from that corresponding to the other eigenvalues. In addition, compared to the maximum feature layer, the other feature layers were distributed near 0. The expressions above can easily lead to the following relation that $F_z = \lambda_z(M_S) \prod_{k=1; k \neq z}^n (\lambda_z(M_S) - \lambda_k(M_S)) T_z$, in which the eigenvalues and their differential products with other eigenvalues cause this phenomenon. This is feasible not only for real symmetric matrices, but also for more general normalized diagonalizable matrices.

In the following, we mainly use the desirable properties of a real symmetric matrix to describe categories from the perspective of attribute and object relations.

3. Relation-based category representation

In the real world, a whole is usually composed of interacting and interrelated parts, and the performance of the combined whole is greater than the sum of the parts, which is the basic principle of system theory [8]. Granular computing is a structured way of thinking [73]. Inspired by system theory and granular computing, the property of the whole does not exist in any of its parts, but arises from the interaction and relations of the parts. We study the category as a whole system. The concept is the abstraction of the category, and the category is the concretization of the concept. We obtained an abstract representation of the concept by establishing a relation space intra-category and salient relation representation. In this section, we represent concretization concepts from two perspectives: the relation structure among attributes and the relation structure among objects. We first propose the attribute and object relation matrices, and then use a salient feature space to represent the attribute relations and object relations. We use the Iris dataset as an example to describe different representational concepts, and verify consistency and second-order isomorphism. The primary framework of the proposed method is illustrated in Fig. 1.

3.1. The representation of relation

In this section, we construct attribute-based relation and object-based relation respectively.

• Attribute-based Relation

Attributes are properties in which certain semantics are shared among different objects. Information is related to and shared among multiple attribute variables. Using the relationship between attributes, we can obtain more meaningful structured information about the attributes. Some attribute relations depend on the given objects, whereas others are independent of the given objects [66]. Recently, research on attribute relation has attracted considerable attention in various fields. Siddiquie et al. [52] demonstrated that pairwise associations between different attributes can be used to achieve better performance. Mutual information [58] is a representative method for describing variable associations that can describe the nonlinear dependence relations between two variables: symmetric, nonlinear, and nonnegative. To this end, we used mutual information to obtain a representation of relations based on attributes.

In this subsection, assuming that a category has m attributes, we treat attributes $A_1, A_2, \dots, A_m$ as m random variables in a category. Mutual information is the reduction of uncertainty of A_i given knowledge of A_j , and is a measure of the interdependence of random variables, considering two random variables A_i and A_j , mutual information $MI(A_i; A_j)$ can be expressed as the relative entropy between the joint distribution $p(a_i, a_j)$ of the random variables A_i and A_j and the product of marginal distributions $p(a_i)$ and $p(a_j)$ of A_i and A_j , i.e.

$$MI(A_i; A_j) = \sum_{a_i \in A_i} \sum_{a_j \in A_j} p(a_i, a_j) \log \frac{p(a_i, a_j)}{p(a_i)p(a_j)}. \quad (12)$$

Mutual information is symmetrical, that is $MI(A_i; A_j) = MI(A_j; A_i)$. Thus, we obtained a relation-symmetric matrix representation of the categories. We denote the mutual information matrix between the attributes as

$$MI_A = \begin{bmatrix} MI(A_1; A_1) & MI(A_1; A_2) & \dots & MI(A_1; A_m) \\ MI(A_2; A_1) & MI(A_2; A_2) & \dots & MI(A_2; A_m) \\ \vdots & \vdots & \ddots & \vdots \\ MI(A_m; A_1) & MI(A_m; A_2) & \dots & MI(A_m; A_m) \end{bmatrix}$$

• Object-based Relation

Like attribute relation, we establish the relation structure among objects based on the similarity relation among them. In order to construct the spatial relation structure of objects, which contains both numerical data and categorical data, Qian et al. proposed a spatial structure representation method for categorical data [43], which provided a data-representation scheme of objects, this representation is made up of the probability of similarity between objects, which can map a categorical dataset to a numeric type in Euclidean space, for numerical data we use the normalized absolute value distance representation. The new representation of objects obtained by the data-representation scheme is denoted by $O_1, O_2, \dots, O_n$, and then similarity relation matrix MS is obtained according to the distance

$$MS(O_i, O_j) = 1 - \sqrt{2(1 - \cos(O_i, O_j))}, \quad (13)$$

where the closer objects O_i and O_j are, the greater the value of $MS(O_i, O_j)$ is, and $MS(O_i, O_j) = MS(O_j, O_i)$. We denote the spatial relation matrix between objects as

Table 1

A data table with relations of attributes.

Attributes	A_1	A_2	$\dots$	A_m
A_1	$R_{A_{11}}$	$R_{A_{12}}$	$\dots$	$R_{A_{1m}}$
A_2	$R_{A_{21}}$	$R_{A_{22}}$	$\dots$	$R_{A_{2m}}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
A_m	$R_{A_{m1}}$	$R_{A_{m2}}$	$\dots$	$R_{A_{mm}}$

$$MS_O = \begin{bmatrix} MS(O_1; O_1) & MS(O_1; O_2) & \dots & MS(O_1; O_n) \\ MS(O_2; O_1) & MS(O_2; O_2) & \dots & MS(O_2; O_n) \\ \vdots & \vdots & \ddots & \vdots \\ MS(O_n; O_1) & MS(O_n; O_2) & \dots & MS(O_n; O_n) \end{bmatrix}$$

We now have an attribute-based relation matrix MI_A and object-based relation matrix MS_O , both of which are symmetric. In the next subsection, we use the salient feature space of the symmetric matrix as outlined in Section 2 to achieve the salient representation of the category based on relation.

3.2. The salient representation of category based on relation

Inspired by philosophy and cognitive psychology, Li et al. [34] proposed granular concepts based on attributes and object cognitive operators, which can be regarded as a type of acquisition of salient concepts, and achieve the acquisition of salient concepts through dynamic incremental correlation when additional information was provided. Based on our research on symmetric matrices in Section 2 and inspired by the idea of saliency, given the relations based on attributes and relations based on objects, we use the salient feature space to propose a salient representation of categories based on attribute relations and a salient representation of categories based on object relations.

• Salient representation based on attribute relation

Category representation based on the relations of attributes aims to use salient relations among the attributes of a category to describe the category, while the salient relations obtained simultaneously can distinguish different categories.

First, we use a mutual information matrix MI_A to represent the nonlinear statistical relations among the attributes of the category, and then obtain the salient representation of the mutual information relations. In particular, the matrix $\mathcal{F}(MI_A)$ based on attribute relations is used to represent the salient relation R_A among attributes, that is

$$R_A = \frac{1}{\alpha} \mathcal{F}(MI_A) = \frac{1}{\alpha} \sum_{i=1}^m \lambda_i (MI_A) \text{adj}(\lambda_i (MI_A) I_n - MI_A), \quad (14)$$

where α is a normalized factor, and $\alpha = \frac{1}{K} \sum_{k=1}^K (\frac{1}{m^2} \sum_{i=1}^m \sum_{j=1}^m r_A^k(i, j))$, K is the number of categories, and the attribute-based relation of the k th category $r_A^k = \sum_{i=1}^m \lambda_i (MI_A^k) \text{adj}(\lambda_i (MI_A^k) I_n - MI_A^k)$.

For convenience, we list the data in Table 1 to show the matrix of attribute-based salient relations, which is a real symmetric matrix, i.e. $R_{A_{ij}} = R_{A_{ji}}$.

• Salient representation based on object relation

Category representation based on the relations of objects aims to use the salient relations among the objects of a category to describe the category, while the salient relations obtained simultaneously can distinguish different categories.

First, we use a relation matrix MS to represent the spatial similarity relations among objects of the category, and then obtain the salient representation of the spatial similarity relations. In particular, the matrix $\mathcal{F}(MS_O)$ based on object relations is used to represent the salient relation R_O among objects, that is

$$R_O = \frac{1}{\beta} \mathcal{F}(MS_O) = \frac{1}{\beta} \sum_{i=1}^m \lambda_i (MS_O) \text{adj}(\lambda_i (MS_O) I_n - MS_O). \quad (15)$$

Similar to attribute, β is a normalized factor, and $\beta = \frac{1}{K} \sum_{k=1}^K (\frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n r_O^k(i, j))$, K is the number of categories, n is the number of objects, and the object-based relation of the k th category $r_O^k = \sum_{i=1}^n \lambda_i (MS_O^k) \text{adj}(\lambda_i (MS_O^k) I_n - MS_O^k)$.

For convenience, we list the data in Table 2 to show the matrix of object-based salient relations, which is a real symmetric matrix, i.e. $R_{O_{ij}} = R_{O_{ji}}$.

We now have an attribute-based salient relation matrix R_A and an object-based salient relation matrix R_O , both of which are symmetric matrices, and are representations of a category from different perspectives. In the next subsection, we focus on the role of such representations in the description of the near-far relations among concepts.

Table 2
A data table with relations of objects.

Objects	O_1	O_2	$\dots$	O_n
O_1	$R_{O_{11}}$	$R_{O_{12}}$	$\dots$	$R_{O_{1n}}$
O_2	$R_{O_{21}}$	$R_{O_{22}}$	$\dots$	$R_{O_{2n}}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
O_m	$R_{O_{m1}}$	$R_{O_{m2}}$	$\dots$	$R_{O_{mn}}$

3.3. The spatial near-far relations among concepts

From the perspective of system theory, the relation structure of a category is representative of this category to a certain extent, and the corresponding relation structures of different categories should be different. Categories can be understood as a group of entities of a concept, and different categories of the same thing can be understood as concretization concepts. We attempt to describe the corresponding concept by extracting the representative structure of a category. In this subsection, we understand the salient relation of a category as an abstract description of the corresponding concept. We first analyze the feasibility of the salient relation representation for the description of concepts, and then find that the salient relation representation can describe certain near-far relations among concepts by comparison, and finally propose the second-order isomorphism verification method.

In this subsection, we discuss the effect of salient representation on concept abstraction, that is, whether the salient representation can describe concepts and their near-far relations. This is discussed in terms of three aspects: consistency, near-far relations, and second-order isomorphism.

3.3.1. Consistency of salient relation representation

To illustrate the validity and consistency of salient relation representation for category description, we conducted extensive experiments on real data sets and found that the categories represented by the original relations cannot consistently distinguish between pairwise attributes, whereas the categories represented by the salient relations show consistent distinctions like the category labels and, on this basis, present consistent near-far relations. This consistency is mainly reflected in the fact that all relations between pairwise attributes show a consistent distinction. However, the salient relation representation of a category further reflects the near-far relations among them, therefore, we abstract it and attempt to represent the corresponding concept of this category. We illustrate this using a real data set below.

We consider the relations among the attributes of Iris data set as an example, which contains 3 categories, each of which refers to a type of iris plant and has 50 instances and 4 attributes. Fig. 3 shows the consistency of salient relation representation by comparing pairwise attribute pairs among labels, mutual information relations, salient relations of different categories respectively, where the blue scatters represent the category Iris-setosa, the red scatters represent the category Iris-versicolor, and the yellow scatters represent the category Iris-virginica. Fig. 3(a)–(f) visualize the mutual information relation MI of the pairwise attributes of different categories, Fig. 3(g)–(l) visualize the salient relation representation $R_A = \frac{1}{\alpha} \mathcal{F}(MI_A)$ of the pairwise attributes of different categories, and Fig. 3(m)–(r) visualize the labels of the pairwise attributes of different categories.

By comparing the three representations of mutual information, salient relations and category labels, we find that the salient relation representation of different attribute pairs is consistent among different categories, and it is consistent for the description of near-far relations between different attribute pairs among different categories, as shown in Fig. 3. In Fig. 3(m)–(r), it is clear that the representation of category labels is consistent in relation to different attribute pairs, and the order among the categories is Iris-virginica (yellow), Iris-versicolor (red), and Iris-setosa (blue). However, in Fig. 3(a)–(f), the relations between attribute pairs directly represented by mutual information cannot achieve a consistent distinction or near-far relations among different categories. For example, Fig. 3(a)(f) and Fig. 3(b)–(e) show different orders and near-far relations, respectively. What is different is that after the salient relation representation in Fig. 3(g)–(l), we not only achieve the same consistency as the category label representation and clear distinction, but also reflect the near-far relations among categories.

In summary, the homogeneity of category labels has no distinction in terms of near-far relations among categories, and the original relation does not have consistency for pairwise attribute pair representations of different categories, whereas salient relations reflect near-far relations, which are consistent for both attribute-based and object-based representations.

3.3.2. Salient representation to describe the near-far relations among concepts

To further illustrate the significance of the salient relation representation for concept abstraction and describing near-far relations among concepts, we conducted experiments to compare the difference between salient and non-salient relations in terms of attribute and object relations in globe. Subsection 3.1 respectively established the relation structures based on attributes and objects, which we call the original relation representation. Based on the relations in Subsection 3.1, the salient relation representation in Subsection 3.2 is obtained using the salient feature space. In this subsection, we compare the salient relations with the original relations for different categories from the perspective of attributes and objects, and find that the salient relation representation has a near-far distinction in the description of concepts.

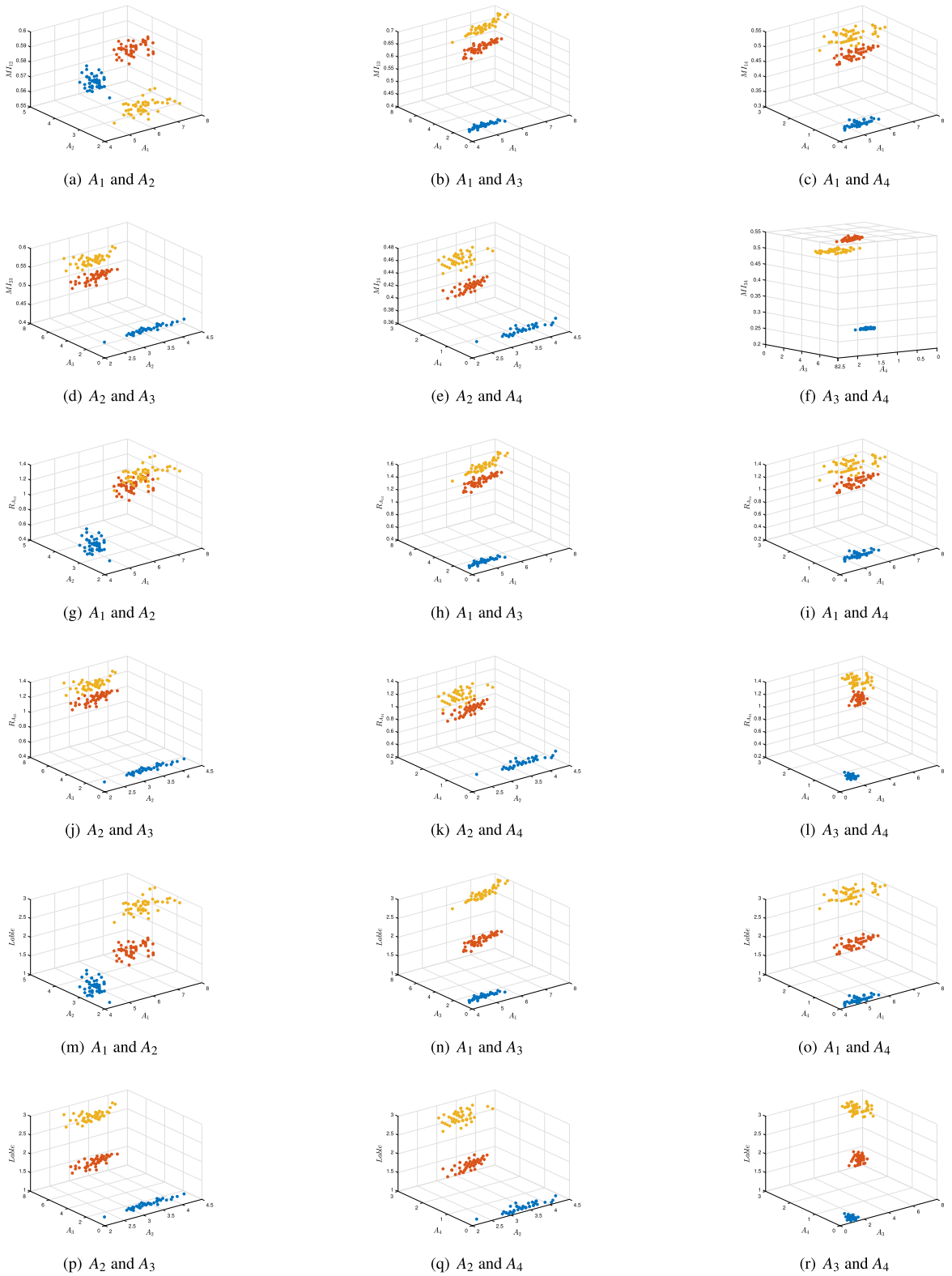


Fig. 3. Consistency of salient relation representation. (For interpretation of the color(s) in the figure(s), the reader is referred to the web version of this article.)

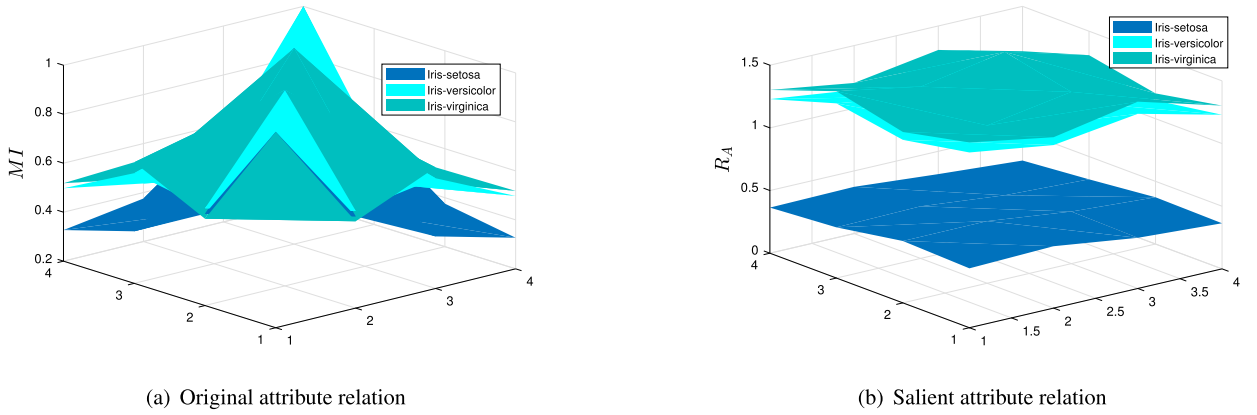


Fig. 4. Difference between original attribute relation and salient attribute relation.

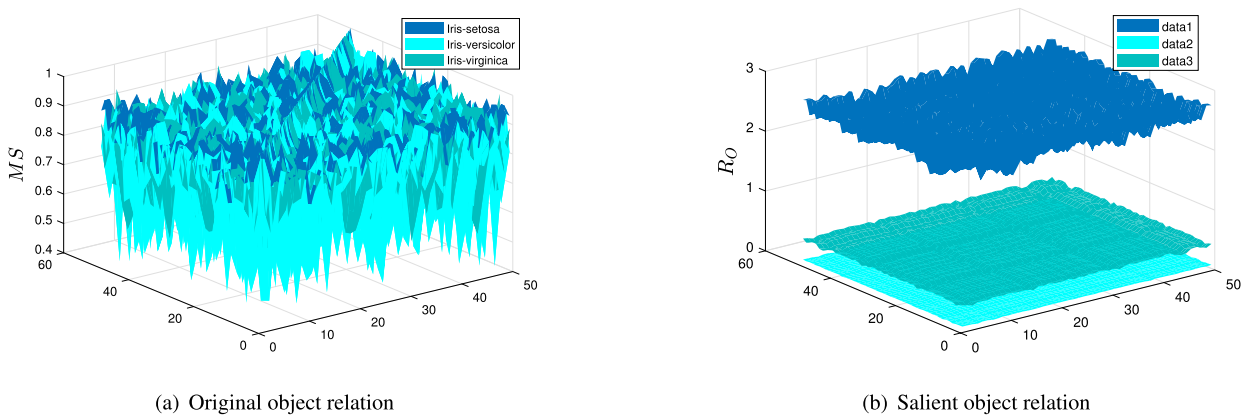


Fig. 5. Difference between original object relation and salient object relation.

In this subsection, we consider the Iris data set as an example. First, we visualized the original and salient relations for different categories based on attributes, as shown in Fig. 4, and then visualized the original and salient relations for different categories based on the objects, as shown in Fig. 5.

Fig. 4 compares the original relation representation and salient relation representation based on attributes, reflecting the advantages of salient relation representation in concept abstraction. Fig. 4(a) shows the mutual information relations MI calculated using Eq. (12) among attributes of different categories in the Iris data set, as can be seen from the figure, the mutual information relation structures of different categories are mixed, and there is no clear distinction among them. However, in Fig. 4(b), the salient relations R_A calculated according to Eq. (14) are visualized, and it can be observed that there is an obvious hierarchical structure among different categories. The original relation structure among attributes can not distinguish multiple categories, and after salient representation, different categories have clear distinctions and present certain near-far relations.

Similarly, we performed the same analysis for object relations. Fig. 5 compares the original relation representation and the salient relation representation based on objects, reflecting the advantages of salient relation representation in concept abstraction. Fig. 5(a) shows the spatial relations MS calculated using Eq. (13) among objects of different categories in the Iris data set, as can be seen from the figure, the relation structures of different categories are mixed, and there is no clear distinction among them. However, in Fig. 5(b), the salient relations R_O calculated according to Eq. (15) are visualized, and it can be observed that there is an obvious hierarchical structure among different categories. The original relation structure among objects can not distinguish multiple categories, and after salient representation, different categories have clear distinctions and present certain near-far relations.

According to the above analysis, we conclude that the salient relation has a representative effect and abstraction on the concepts corresponding to the category, and the salient relation structure can show the spatial near-far relations among concepts to some extent compared with the original relations.

3.3.3. Second-order isomorphism of concept

We take the salient representation of the relation structure of intra-category as the abstraction of the corresponding concept from the perspectives of intent (attributes) and extent (objects) respectively, and through the qualitative analysis

Table 3
Data sets description.

Number	Data sets	Instances	Number of attributes	Number of classes	Download
1	BUPA	345	6	2	KEEL
2	Contraceptive Method Choice	1473	9	3	UCI
3	Iris	150	4	3	UCI
4	Heart	270	13	2	KEEL
5	Breast Cancer	286	9	2	UCI
6	Mushroom	8124	22	2	UCI
7	Seeds	210	7	3	UCI
8	Saheart	462	9	2	KEEL
9	Page-blocks	5472	10	5	KEEL
10	Wine	178	13	3	UCI
11	Waveform Database Generator	5000	21	3	UCI
12	Speaker Accent Recognition	329	12	6	UCI

in the previous two subsections, we find that such a representation can distinguish the spatial distance between concepts. In the following, we demonstrate that salient representation can describe the spatial near-far relations among concepts through a quantitative measure.

In cognitive psychology, although there is no structural similarity between the internal representation of an individual and the corresponding external object, similar things are similar in our internal representations or mental intentions and external representations, which is called second-order isomorphism [48,45]. For example, although the inner representation of a square is not necessarily square, it has a closer relationship with the representation of a quadrilateral than with a triangle or circle.

A concept can be characterized by its intent and extent, while categories are concrete representations of the concept. In this study, intent is represented by the attributes of a category, and extent is represented by the objects of the same category. According to the view of system theory, the salient representation of relation structure is strongly representative of categories, which has been analyzed qualitatively in the previous subsections. Accordingly, we provide an abstract representation of concepts from the perspective of intent and extent relations. In this subsection, we propose a quantitative measure RS of the second-order isomorphism to measure the validity of this salient representation. We refer to the structural similarity measure [68] to construct the calculation method for the relational similarity matrix RS among the concepts described by the salient relations, that is

$$RS(i, j) = \frac{C}{\|R_i - R_j\|_{L_1} + C} \cdot \frac{(2\mu_i\mu_j + C_1)(2\sigma_i\sigma_j + C_2)}{(\mu_i^2 + \mu_j^2 + C_1)(\sigma_i^2 + \sigma_j^2 + C_2)}, \quad (16)$$

where $\frac{C}{\|R_i - R_j\|_{L_1} + C}$ measures the similarity of the relations between the two categories in size and ranges from 0 to 1, and $\frac{(2\mu_i\mu_j + C_1)(2\sigma_i\sigma_j + C_2)}{(\mu_i^2 + \mu_j^2 + C_1)(\sigma_i^2 + \sigma_j^2 + C_2)}$ measures the similarity of the two category relations in terms of structural similarity, ranging from -1 to 1 . μ_i and μ_j represent the means of the relation matrices of i th category R_i and j th category R_j respectively, σ_i and σ_j represent the variances of the relation matrices R_i of i th category and R_j j th category respectively. And C , C_1 , and C_2 are constants, C ensures the range of the first factor in Eq. (16), C_1 and C_2 make sure that the denominator of the second factor in Eq. (16) is not 0. Thus, RS can measure the similarity of the relation between two categories more comprehensively and ranges from -1 to 1 . The second-order isomorphism can then be computed as the Spearman correlation between the upper diagonal portion of the two similarity matrices of attribute-based and object-based, where the mapping determines the order of concepts in the matrices. The higher Spearman correlation, the stronger the second-order isomorphism.

4. Results and experimental analysis

We demonstrated that the salient relation representation can abstract concepts and describe spatial near-far relations among them through preliminary qualitative analysis in Section 3. To further demonstrate the effect of the near-far relations description of salient relation representation, we conducted a series of experiments on 12 representative data sets, as shown in Table 3, including numerical data sets, categorical data sets and mixed data sets. In this section, we conduct a salient relation representation experiment of categories from the perspective of attributes and objects, based on the above experiments, we use more than three categories of data to quantitatively verify the existence of second-order isomorphism of the intent and extent of the concept. We explain and analyze the experimental results, and discuss the reasons for the experimental phenomena as well as the differences and connections between the proposed and existing methods.

4.1. Relations-based category representation experiments

In unsupervised machine learning, a category is typically represented by a prototype or cluster center. In supervised machine learning, labels are used to represent categories. Such representations, particularly labels, have no bias and ignore

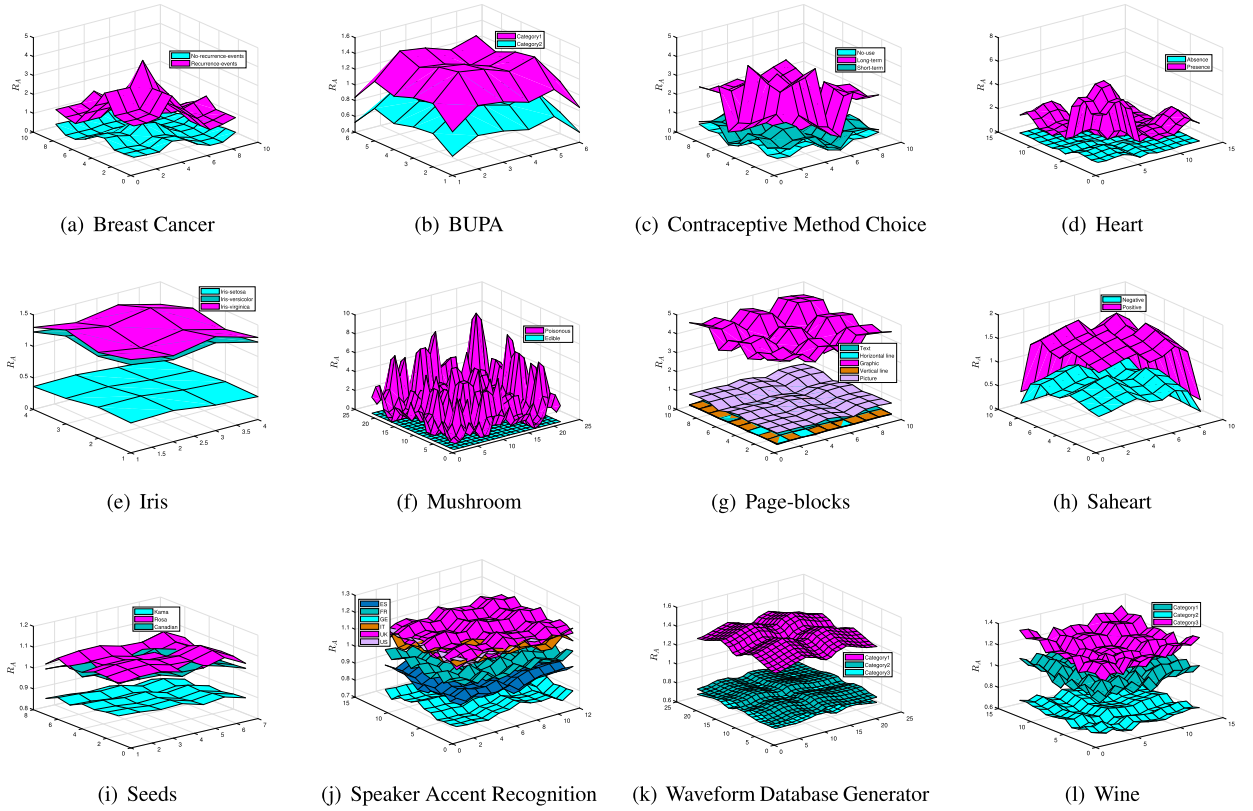


Fig. 6. Category representation based on relations of attributes.

the near-far relations among the categories. In this experimental analysis, we aimed to verify the advantages of the salient relation representation of categories from two perspectives: attribute-based and object-based.

• Attribute-based category representation

In this experiment, for each given data set, first, we use Eq. (12) in Subsection 3.1 to calculate the mutual information relations between pairwise attribute pairs and obtain a symmetric mutual information relation matrix. Subsequently, using the salient feature space of the symmetric matrix, Eq. (14) in Subsection 3.2 is used to realize the salient relation representation of the relation matrix obtained above, so as to realize the conceptual abstract representation of categories based on attributes. We represented different categories of each data set, and the experimental results are shown in Fig. 6.

Fig. 6(a)–(l) respectively correspond to attribute-based category representation of 12 data sets, which can be seen from the figure, the salient relation representation based on attributes can effectively distinguish among different categories, that is, the different categories of each dataset can be clearly distinguished by attribute-based salient relation representation. On this basis, the similarities among different categories in the same data set are obviously different, and different near-far relations are presented, which also reflects that the salient relation representation has a good representative and abstract effect on the categories.

• Object-based category representation

Similar to attribute-based category representation, for verifying the efficiency of object-based salient relation representation of categories, first, we randomly downsample 30 samples for each category, and re-represent the object data in Euclidean space using the spatial representation method proposed by [43] for categorical data, and using the normalized absolute value distance representation for numerical data. Then, we use Eq. (13) in Subsection 3.1 to calculate the relations between pairwise object pairs and obtain the symmetric spatial relation matrix. Then, using the salient feature space of symmetric matrix, Eq. (15) in Subsection 3.2 is used to realize the salient relation representation of the relation matrix obtained above, so as to realize the conceptual abstract representation of categories based on objects. We represented different categories of each data set, and the experimental results are shown in Fig. 7.

Fig. 7(a)–(l) respectively correspond to object-based category representation of 12 data sets, which can be seen from the figure, the salient relation representation based on objects can effectively distinguish among different categories, that is, we observe the same results as attribute-based representation of category that different categories of each dataset can be clearly distinguished by object-based salient relation representation. On this basis, the similarities among different categories in

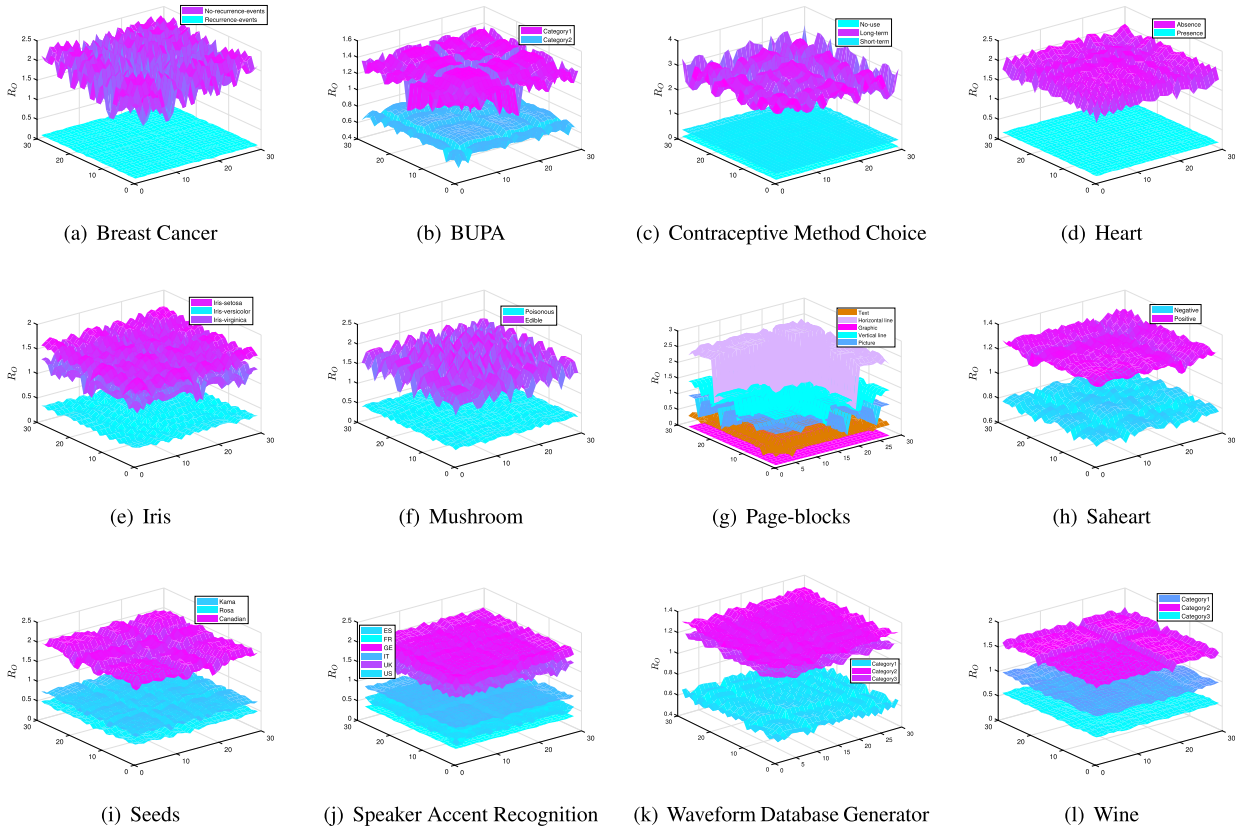


Fig. 7. Category representation based on relations of objects.

the same data set are obviously different, and different near-far relations are presented, which also reflects that the salient relation representation based on object has a good representative and abstract effect on the categories.

In summary, this study implements category representation based on attribute and object relations, which can not only effectively distinguish different categories, but also describe the spatial near-far relations among categories.

4.2. Second-order isomorphism verification experiment

Through the experiments described in the previous section, we found that salient relation representation has both representativeness and an abstraction effect on categories. Therefore, we used a quantitative measure to experimentally verify and analyze whether such a representation can make similar things more similar.

Relations among internal attribute structures and relations among external object structures of different categories should have similar ordering relationships, the so-called second-order isomorphism. For a data set, we first calculated two relation similarity matrices among categories based on attributes and objects respectively according to Eq. (16), which measures the similarity of categories in scale and structure. We then calculated the Spearman correlation of the two matrices to verify second-order isomorphism. More importantly, we verified the validity of salient relation representation through second-order isomorphism between intent and extent of concept, and vice versa.

To better reflect the second-order isomorphism of the salient relation representation, we selected the data sets with more than two categories for the experiment and compared them with the second-order isomorphism of the original relation. In this experiment, we set $C = 100$, $C_1 = 0.001$ and $C_2 = 0.003$ in Eq. (16). Fig. 8(a)–(f) show the Spearman correlations between attribute relations and object relations of the original relation representation and the salient relation representation respectively for three data sets (Iris, Wine, and Speaker Accent Recognition). Experimental results of more data sets are shown in Table 4.

In Fig. 8(a), the upper subfigure measures the similarity of the attribute-based original relations among categories of the Iris dataset using Eq. (16), which is used to represent the near-far relations among categories based on attributes, and the lower subfigure measures the similarity of the object-based original relations among categories of the Iris dataset using Eq. (16), which is used to represent the near-far relations among categories based on objects. The consistency of the upper and lower near-far relations was measured using Spearman correlation. Fig. 8(b)(c) are the same as those shown in (a). In Fig. 8(d), the upper subfigure measures the similarity of the attribute-based salient relations among categories of the Iris



Fig. 8. The second-order isomorphism of concept.

dataset using Eq. (16), which is used to represent the near-far relations among concepts based on attributes (we use the salient relation representation to represent the concept abstraction of the corresponding category from the perspective of attributes), and the lower subfigure measures the similarity of the object-based salient relations among concepts of the Iris dataset using Eq. (16), which is used to represent the near-far relations among categories based on objects (we use the salient relation representation to represent the concept abstraction of the corresponding category from the perspective of objects). The consistency of the upper and lower near-far relations was measured using Spearman correlation. Fig. 8(e)(f) are identical to those shown in (d).

Table 4
Spearman correlation coefficient.

Data sets	Spearman Corr of original relation	Spearman Corr of salient relation
Contraceptive Method Choice	0.9355	1
Iris	0.8065	1
Seeds	0.9355	1
Page-blocks	0.9259	0.9259
Wine	0.7419	0.9355
Waveform Database Generator	0.9355	0.9355
Speaker Accent Recognition	0.4086	0.7847

By comparing the Spearman correlations between (a) and (d), (b) and (e), and (c) and (f) in Fig. 8, it is easy to note that the salient relation representation can better represent the spatial near-far relations, make concepts that are externally similar internally similar, and better reflect second-order isomorphism than the original relation representation. It can also be seen from the comparison data in Table 4 that the salient relation representation reflects the second-order isomorphism better than the original relation representation.

4.3. Discussion

We demonstrated the representational effect of salient relation representation on categories qualitatively and quantitatively through the description of near-far relations among concepts and the second-order isomorphism of concepts experimentally. In this subsection, the reasons for these phenomena are analyzed, and the main differences and connections between the proposed and existing methods are discussed.

From the relation matrix among the attributes of category and the relation matrix among the objects of category, salient relations are obtained to describe and understand the concepts, which can distinguish the categories better, as shown in Fig. 6 and Fig. 7. Salient relation representation is representative of concepts in terms of spatial near-far relations, and can achieve the goal of externally similar concepts being internally similar. The main reasons for this are as follows:

- From the perspective of the ideas adopted in the article, (1) Inspired by system theory, the whole is greater than the sum of its parts, and the representation of the relational structure can reflect the whole structure to some extent. (2) Inspired by the multiple perspectives of granular computing, it is reasonable to express categories based on the intent and extent of concepts.
- From the perspective of the theoretical method, (1) The symmetric relation between multiple variables can be expressed by a symmetric matrix. Therefore, we first research a good representation of the symmetric matrix. Based on this, we propose the salient feature space, and try to put forward the salient relation representation method. (2) A variable that is useless by itself can significantly improve the performance when used with other variables, and two variables that are useless still provide good class separability when used together. The method presented in this article relates to all variables. (3) Salient relation representation means that the factor of the eigenvalue difference product is removed to magnify the effect of larger eigenvalues while retaining all the feature information. Salient representation is realized from the perspective of global relations.
- From the perspective of designing experiment: step by step, we demonstrated the effect of the salient representation on describing the spatial near-far relations among concepts from three perspectives of consistency, near-far relations, and second-order isomorphism. We not only considered the attribute relations and object relationship, but also compared with the original relations (as shown in Fig. 3, Fig. 4, Fig. 5, Fig. 8, and Table 4).

We analyzed the main differences and connections with existing methods. (1) In contrast to many reduction and compression methods, all the relation information was considered in this study for a salient representation. (2) Different from simply considering relations, this study implemented a salient relation representation based on relations. (3) In contrast to previous concept abstraction, this study begins by studying the representation of symmetric matrices, uses salient structural relations to abstract and represent categories and verifies the second-order isomorphism of concepts.

5. Conclusion and future work

In this study, the idea of human cognition and understanding of concepts is introduced into the data analysis process, and we find that the spatial near-far relations among concepts are considered to understand conceptual categories. A novel relation-based method for understanding different concepts was proposed.

The following conclusions were obtained through theoretical analysis and experimental verification: attribute-based and object-based relations were proposed to describe categories; to better distinguish different concepts in machine learning, a salient relation representation method of categories is proposed that can distinguish the spatial near-far relation among concepts; the validity of salient relation representation is obtained through the second-order isomorphism of conceptual intent and extent. This paper focuses primarily on the understanding and description of concepts based on symmetric

relations, but there are more asymmetric relations in the real world. The salient relation representation method is valid for general diagonalizable matrices, and the study of these asymmetric relations is also a key issue for us to solve a series of problems such as concept understanding, relationship establishment, and reasoning in concept learning. In the future, we will conduct in-depth research on the structural features of the asymmetric relations of concepts. Finally, and most importantly, salient relation representation has the ability to learn about the distance among categories, which can give specific instructions to the machine in the process of concept learning. Thus, it may promote the concept learning based on granular computing, improve machine learning performance, and enlighten the intelligent reasoning of concepts.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgement

This study was supported by National Key Research and Development Program of China (No. 2021ZD0112400), Key Program of the National Natural Science Foundation of China (No. 62136005), Key R&D Program (International Science and Technology Cooperation Project) of Shanxi Province, China (No. 201903D421003), the 1331 Engineering Project of Shanxi Province, China, the Natural Science Foundation of Shanxi Province, China (Nos. 20210302124549, 20210302123455, 20210302124556), the Open Project Program of the Key Laboratory of Embedded System and Service Computing of Ministry of Education (Tongji University), ESSCKF 2021-04, and Scientific and Technological Innovation Programs of Higher Education Institutions in Shanxi (No. 2021L286).

References

- [1] D. Angluin, C.H. Smith, *A Survey of Inductive Inference: Theory and Methods*, Yale University, Department of Computer Science, 1982.
- [2] A. Banerjee, S. Merugu, I.S. Dhillon, J. Ghosh, J. Lafferty, Clustering with Bregman divergences, *J. Mach. Learn. Res.* 6 (10) (2005).
- [3] A. Bargiela, W. Pedrycz, Granular computing, in: *Handbook on Computational Intelligence: Volume 1: Fuzzy Logic, Systems, Artificial Neural Networks, and Learning Systems*, World Scientific, 2016, pp. 43–66.
- [4] A. Barr, E.A. Feigenbaum, *The Handbook of Artificial Intelligence*, Pitman, 1982.
- [5] L. Breiman, J.H. Friedman, R.A. Olshen, C.J. Stone, *Classification and Regression Trees*, Routledge, 2017.
- [6] C.E. Brodley, P.E. Utgoff, Multivariate decision trees, *Mach. Learn.* 19 (1) (1995) 45–77.
- [7] C.J. Burges, A tutorial on support vector machines for pattern recognition, *Data Min. Knowl. Discov.* 2 (2) (1998) 121–167.
- [8] F. Capra, *The Web of Life: A New Scientific Understanding of Living Systems*, Anchor, 1997.
- [9] M. Caron, P. Bojanowski, A. Joulin, M. Douze, Deep clustering for unsupervised learning of visual features, in: *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018, pp. 132–149.
- [10] N. Cristianini, J. Shawe-Taylor, et al., *An Introduction to Support Vector Machines and Other Kernel-Based Learning Methods*, Cambridge University Press, 2000.
- [11] P. Denton, S. Parke, T. Tao, X. Zhang, Eigenvectors from eigenvalues: a survey of a basic identity in linear algebra, *Bull. Am. Math. Soc.* (2021).
- [12] R.O. Duda, P.E. Hart, et al., *Pattern Classification and Scene Analysis*, vol. 3, Wiley, New York, 1973.
- [13] R. Feinman, B.M. Lake, Generating new concepts with hybrid neuro-symbolic models, *arXiv preprint*, arXiv:2003.08978, 2020.
- [14] B. Ganter, R. Wille, *Formal Concept Analysis: Mathematical Foundations*, Springer-Verlag New York, Inc, 1999.
- [15] D. Gentner, K.J. Kurtz, *Relational Categories*, 2005.
- [16] R.L. Goldstone, J.Y. Son, *Similarity*, Oxford University Press, 2012.
- [17] N.D. Goodman, J.B. Tenenbaum, J. Feldman, T.L. Griffiths, A rational analysis of rule-based concept learning, *Cogn. Sci.* 32 (1) (2008) 108–154.
- [18] H. Guo, S.B. Gelfand, Classification trees with neural network feature extraction, in: *Proceedings 1992 IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, IEEE Computer Society, 1992, pp. 183–184.
- [19] I. Guyon, A. Elisseeff, An introduction to variable and feature selection, *J. Mach. Learn. Res.* 3 (Mar 2003) 1157–1182.
- [20] J.A. Hartigan, M.A. Wong, Algorithm as 136: a k-means clustering algorithm, *J. R. Stat. Soc., Ser. C, Appl. Stat.* 28 (1) (1979) 100–108.
- [21] C. Huang, J. Li, C. Mei, W.-Z. Wu, Three-way concept learning based on cognitive operators: an information fusion viewpoint, *Int. J. Approx. Reason.* 83 (2017) 218–242.
- [22] S. Huang, M. Elhoseiny, A. Elgammal, D. Yang, Learning hypergraph-regularized attribute predictors, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2015, pp. 409–417.
- [23] S.M. Kazemi, D. Poole, Relnn: a deep neural model for relational learning, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, 2018.
- [24] C. Kemp, J.B. Tenenbaum, The discovery of structural form, *Proc. Natl. Acad. Sci.* 105 (31) (2008) 10687–10692.
- [25] C. Kemp, J.B. Tenenbaum, T.L. Griffiths, T. Yamada, N. Ueda, Learning systems of concepts with an infinite relational model, in: *AAAI*, vol. 3, 2006, p. 5.
- [26] A. Kovashka, K. Grauman, Attributes for image retrieval, in: *Visual Attributes*, Springer, 2017, pp. 89–117.
- [27] A. Kovashka, S. Vijayanarasimhan, K. Grauman, Actively selecting annotations among objects and attributes, in: *2011 International Conference on Computer Vision*, IEEE, 2011, pp. 1403–1410.
- [28] B. Lake, R. Salakhutdinov, J. Gross, J. Tenenbaum, One shot learning of simple visual concepts, in: *Proceedings of the Annual Meeting of the Cognitive Science Society*, vol. 33, 2011.
- [29] B.M. Lake, N.D. Lawrence, J.B. Tenenbaum, The emergence of organizing structure in conceptual representation, *Cogn. Sci.* 42 (2018) 809–832.

- [30] B.M. Lake, R. Salakhutdinov, J.B. Tenenbaum, Human-level concept learning through probabilistic program induction, *Science* 350 (6266) (2015) 1332–1338.
- [31] J. Li, Z. He, Q. Zhu, An entropy-based weighted concept lattice for merging multi-source geo-ontologies, *Entropy* 15 (6) (2013) 2303–2318.
- [32] J. Li, C. Huang, J. Qi, Y. Qian, W. Liu, Three-way cognitive concept learning via multi-granularity, *Inf. Sci.* 378 (2017) 244–263.
- [33] J. Li, C. Huang, W. Xu, Y. Qian, W. Liu, Cognitive concept learning via granular computing for big data, in: 2015 International Conference on Machine Learning and Cybernetics (ICMLC), vol. 1, IEEE, 2015, pp. 289–294.
- [34] J. Li, C. Mei, W. Xu, Y. Qian, Concept learning via granular computing: a cognitive viewpoint, *Inf. Sci.* 298 (2015) 447–467.
- [35] Y. Li, P. Hu, Z. Liu, D. Peng, J.T. Zhou, X. Peng, Contrastive clustering, in: AAAI Conference on Artificial Intelligence, 2021.
- [36] M. Liu, D. Zhang, S. Chen, Attribute relation learning for zero-shot classification, *Neurocomputing* 139 (2014) 34–46.
- [37] J. Lu, J. Li, Z. Yan, F. Mei, C. Zhang, Attribute-based synthetic network (abs-net): learning more from pseudo feature representations, *Pattern Recognit.* 80 (2018) 129–142.
- [38] J. Mao, C. Gan, P. Kohli, J.B. Tenenbaum, J. Wu, The neuro-symbolic concept learner: interpreting scenes, words, and sentences from natural supervision, *arXiv preprint, arXiv:1904.12584*, 2019.
- [39] Y. Mi, Y. Shi, J. Li, W. Liu, M. Yan, Fuzzy-based concept learning method: exploiting data with fuzzy conceptual clustering, *IEEE Trans. Cybern.* (2020).
- [40] R. Michalski, J. Carbonell, T. Mitchell, *Machine Learning: An Artificial Intelligence Approach*, 1983.
- [41] J. Niu, C. Huang, J. Li, M. Fan, Parallel computing techniques for concept-cognitive learning based on granular computing, *Int. J. Mach. Learn. Cybern.* 9 (11) (2018) 1785–1805.
- [42] Y. Qian, H. Cheng, J. Wang, J. Liang, W. Pedrycz, C. Dang, Grouping granular structures in human granulation intelligence, *Inf. Sci.* 382 (2017) 150–169.
- [43] Y. Qian, F. Li, J. Liang, B. Liu, C. Dang, Space structure and clustering of categorical data, *IEEE Trans. Neural Netw. Learn. Syst.* 27 (10) (2016) 2047–2059.
- [44] Y. Qian, X. Liang, Q. Wang, J. Liang, B. Liu, A. Skowron, Y. Yao, J. Ma, C. Dang, Local rough set: a solution to rough data analysis in big data, *Int. J. Approx. Reason.* 97 (2018) 38–63.
- [45] B.D. Roads, B.C. Love, Learning as the unsupervised alignment of conceptual systems, *Nat. Mach. Intell.* 2 (1) (2020) 76–82.
- [46] A. Santoro, D. Raposo, D.G. Barrett, M. Malinowski, R. Pascanu, P. Battaglia, T. Lillicrap, A simple neural network module for relational reasoning, *Adv. Neural Inf. Process. Syst.* 30 (2017).
- [47] C.E. Shannon, A mathematical theory of communication, *Bell Syst. Tech. J.* 27 (3) (1948) 379–423.
- [48] R.N. Shepard, S. Chipman, Second-order isomorphism of internal representations: shapes of states, *Cogn. Psychol.* 1 (1) (1970) 1–17.
- [49] Y. Shi, Y. Mi, J. Li, W. Liu, Concept-cognitive learning model for incremental concept learning, *IEEE Trans. Syst. Man Cybern. Syst.* 51 (2) (2018) 809–821.
- [50] Y. Shi, Y. Mi, J. Li, W. Liu, Concurrent concept-cognitive learning model for classification, *Inf. Sci.* 496 (2019) 65–81.
- [51] R. Shivhare, A.K. Cherukuri, J. Li, Establishment of cognitive relations based on cognitive informatics, *Cogn. Comput.* 9 (5) (2017) 721–729.
- [52] B. Siddiquie, R.S. Feris, L.S. Davis, Image ranking and retrieval based on multi-attribute queries, in: *CVPR 2011, IEEE, 2011*, pp. 801–808.
- [53] P.K. Singh, A.K. Cherukuri, J. Li, Concepts reduction in formal concept analysis with fuzzy setting using Shannon entropy, *Int. J. Mach. Learn. Cybern.* 8 (1) (2017) 179–189.
- [54] P.K. Singh, A. Gani, Fuzzy concept lattice reduction using Shannon entropy and Huffman coding, *J. Appl. Non-Class. Log.* 25 (2) (2015) 101–119.
- [55] K. Sumangali, C.A. Kumar, J. Li, Concept compression in formal concept analysis using entropy-based attribute priority, *Appl. Artif. Intell.* 31 (3) (2017) 251–278.
- [56] J.B. Tenenbaum, *Bayesian Modeling of Human Concept Learning*, MIT Press, 1998.
- [57] J.B. Tenenbaum, Rules and similarity in concept learning, *Adv. Neural Inf. Process. Syst.* 12 (2000) 59–65.
- [58] M. Thomas, A.T. Joy, *Elements of Information Theory*, Wiley-Interscience, 2006.
- [59] M.T. Tomlinson, B.C. Love, Relation-based categories are easier to learn than feature-based categories, in: *Proceedings of the Annual Meeting of the Cognitive Science Society*, vol. 29, 2007.
- [60] A. Tversky, Features of similarity, *Psychol. Rev.* 84 (4) (1977) 327.
- [61] L.G. Valiant, A theory of the learnable, *Commun. ACM* 27 (11) (1984) 1134–1142.
- [62] R. Vedantam, A. Szlam, M. Nickel, A. Morcos, B.M. Lake, Curi: a benchmark for productive concept learning under uncertainty, in: *International Conference on Machine Learning, PMLR, 2021*, pp. 10519–10529.
- [63] G. Wang, Dgcc: data-driven granular cognitive computing, *Granul. Comput.* 2 (4) (2017) 343–355.
- [64] G. Wang, C. Xu, D. Li, Generic normal cloud model, *Inf. Sci.* 280 (2014) 1–15.
- [65] G. Wang, Q. Zhang, J. Hu, An overview of granular computing, *CAAI Trans. Intell. Syst.* 2 (6) (2007) 8–26.
- [66] X. Wang, Q. Ji, A unified probabilistic approach modeling relationships between attributes and objects, in: *Proceedings of the IEEE International Conference on Computer Vision, 2013*, pp. 2120–2127.
- [67] Y. Wang, G. Mori, A discriminative latent model of object classes and attributes, in: *European Conference on Computer Vision, Springer, 2010*, pp. 155–168.
- [68] Z. Wang, A.C. Bovik, H.R. Sheikh, E.P. Simoncelli, Image quality assessment: from error visibility to structural similarity, *IEEE Trans. Image Process.* 13 (4) (2004) 600–612.
- [69] L. Wei, L. Cao, J. Qi, W. Zhang, Concept reduction and concept characteristics in formal concept analysis, *Sci. Sin. Inform.* 50 (12) (2020) 1817.
- [70] R. Wille, Restructuring lattice theory: an approach based on hierarchies of concepts, in: *Proceedings of the NATO Advanced Study Institute, 1982*, pp. 445–470.
- [71] S. Yang, Y. Wang, Y. Shi, Z. Fei, Can categories and attributes be learned in a multi-task way?, *IEEE Trans. Multimed.* 21 (12) (2019) 3194–3204.
- [72] J.T. Yao, A.V. Vasilakos, W. Pedrycz, Granular computing: perspectives and challenges, *IEEE Trans. Cybern.* 43 (6) (2013) 1977–1989.
- [73] Y. Yao, Interpreting concept learning in cognitive informatics and granular computing, *IEEE Trans. Syst. Man Cybern., Part B, Cybern.* 39 (4) (2009) 855–866.
- [74] Y. Yao, Symbols-meaning-value (smv) space as a basis for a conceptual model of data science, *Int. J. Approx. Reason.* 144 (2022) 113–128.
- [75] Y. Yao, et al., Granular computing: basic issues and possible solutions, in: *Proceedings of the 5th Joint Conference on Information Sciences*, vol. 1, Citeseer, 2000, pp. 186–189.
- [76] L.A. Zadeh, Toward a theory of fuzzy information granulation and its centrality in human reasoning and fuzzy logic, *Fuzzy Sets Syst.* 90 (2) (1997) 111–127.
- [77] Y. Zhao, J. Li, W. Liu, W. Xu, Cognitive concept learning from incomplete information, *Int. J. Mach. Learn. Cybern.* 8 (1) (2017) 159–170.
- [78] Y. Zhou, B.M. Lake, Flexible compositional learning of structured visual concepts, *arXiv preprint, arXiv:2105.09848*, 2021.